

SQL Code

```

1  /* =====
2      1. CREATE DATABASE AND USE IT
3  ===== */
4  CREATE DATABASE IF NOT EXISTS data_cleaning_students_dataset;
5  USE data_cleaning_students_dataset;
6
7
8  /* =====
9      2. CREATE CLEAN WORKING TABLE (v2)
10     - Include row_num for deduplication
11 ===== */
12 CREATE TABLE student_dataset_v2 AS
13 SELECT *,
14     ROW_NUMBER() OVER (
15         PARTITION BY Student_ID, Name, Age, Gender, City, Class,
16                     Attendance_Percentage, Study_Hours_per_day,
17                     Math_Score, Science_Score, English_Score,
18                     Total_Score, Parent_Education,
19                     Internet_Access, Result
20     ) AS row_num
21 FROM student_dataset_v1;
22
23
24 /* =====
25     3. REMOVE DUPLICATES
26 ===== */
27 DELETE FROM student_dataset_v2
28 WHERE row_num > 1;
29
30 ALTER TABLE student_dataset_v2 DROP COLUMN row_num;
31
32
33 /* =====
34     4. STANDARDIZE STUDENT_ID FORMAT
35 ===== */
36 UPDATE student_dataset_v2
37 SET Student_ID =
38     CONCAT('STU-',
39         LPAD(REGEXP_REPLACE(Student_ID, '[^0-9]', ''), 5, '0'))
40 WHERE Student_ID NOT LIKE 'STU-%';
41
42
43 /* =====
44     5. CLEAN NAME COLUMN
45     - Trim spaces
46     - Proper case (First letter uppercase)
47 ===== */
48 UPDATE student_dataset_v2
49 SET Name = TRIM(Name);
50

```

```

51 UPDATE student_dataset_v2
52 SET Name = CONCAT(
53     UPPER(LEFT(Name, 1)),
54     LOWER(SUBSTRING(Name, 2))
55 );
56
57
58 /* =====
59     6. HANDLE MISSING NAMES
60 ===== */
61 UPDATE student_dataset_v2
62 SET Name = 'UNKNOWN'
63 WHERE Name IS NULL OR Name = '';
64
65
66 /* =====
67     7. CLEAN AGE COLUMN
68     - Remove invalid values
69     - Fix text values
70 ===== */
71 UPDATE student_dataset_v2
72 SET Age = NULL
73 WHERE Age > 18 OR Age < 0;
74
75 UPDATE student_dataset_v2
76 SET Age = 15
77 WHERE LOWER(Age) = 'fifteen';
78
79
80 /* =====
81     8. CLEAN CLASS COLUMN (TEXT → INT)
82 ===== */
83 UPDATE student_dataset_v2
84 SET Class = CASE
85     WHEN LOWER(Class) IN ('zero') THEN 0
86     WHEN LOWER(Class) IN ('one') THEN 1
87     WHEN LOWER(Class) IN ('two') THEN 2
88     WHEN LOWER(Class) IN ('three') THEN 3
89     WHEN LOWER(Class) IN ('four') THEN 4
90     WHEN LOWER(Class) IN ('five') THEN 5
91     WHEN LOWER(Class) IN ('six') THEN 6
92     WHEN LOWER(Class) IN ('seven') THEN 7
93     WHEN LOWER(Class) IN ('eight') THEN 8
94     WHEN LOWER(Class) IN ('nine') THEN 9
95     WHEN LOWER(Class) IN ('ten') THEN 10
96     ELSE CAST(REGEXP_REPLACE(Class, '[^0-9]', '') AS UNSIGNED)
97 END;
98
99
100 /* =====
101     9. CLEAN GENDER COLUMN + MODE IMPUTATION
102 ===== */
103 UPDATE student_dataset_v2
104 SET Gender = CASE

```

```

105     WHEN LOWER(TRIM(Gender)) IN ('f', 'female') THEN 'Female'
106     WHEN LOWER(TRIM(Gender)) IN ('m', 'male') THEN 'Male'
107     WHEN LOWER(TRIM(Gender)) IN ('o', 'other') THEN 'Other'
108     ELSE NULL
109 END;
110
111 -- Fill NULL with MODE (most frequent value)
112 UPDATE student_dataset_v2
113 SET Gender = (
114     SELECT Gender
115     FROM (
116         SELECT Gender
117         FROM student_dataset_v2
118         WHERE Gender IS NOT NULL
119         GROUP BY Gender
120         ORDER BY COUNT(*) DESC
121         LIMIT 1
122     ) t
123 )
124 WHERE Gender IS NULL;
125
126
127 /* =====
128 10. CLEAN CITY COLUMN
129 ===== */
130 UPDATE student_dataset_v2
131 SET City = TRIM(City);
132
133 UPDATE student_dataset_v2
134 SET City = CONCAT(
135     UPPER(LEFT(City, 1)),
136     LOWER(SUBSTRING(City, 2))
137 );
138
139
140 /* Fill missing City with MODE */
141 UPDATE student_dataset_v2
142 SET City = (
143     SELECT City FROM (
144         SELECT City
145         FROM student_dataset_v2
146         GROUP BY City
147         ORDER BY COUNT(*) DESC
148         LIMIT 1
149     ) t
150 )
151 WHERE City IS NULL OR City = '';
152
153
154 /* =====
155 11. ATTENDANCE CLEANING
156 ===== */
157 UPDATE student_dataset_v2
158 SET Attendance_Percentage = NULL

```

```

159 WHERE Attendance_Percentage > 100 OR Attendance_Percentage < 0;
160
161
162 /* Fill missing attendance with MEDIAN (correct approach) */
163 WITH Ordered AS (
164     SELECT Attendance_Percentage,
165            ROW_NUMBER() OVER (ORDER BY Attendance_Percentage) rn,
166            COUNT(*) OVER () cnt
167     FROM student_dataset_v2
168 )
169 UPDATE student_dataset_v2
170 SET Attendance_Percentage = (
171     SELECT AVG(Attendance_Percentage)
172     FROM Ordered
173     WHERE rn IN (FLOOR((cnt+1)/2), CEIL((cnt+1)/2))
174 )
175 WHERE Attendance_Percentage IS NULL;
176
177
178 /* =====
179 12. STUDY HOURS CLEANING
180 ===== */
181 UPDATE student_dataset_v2
182 SET Study_Hours_per_day = NULL
183 WHERE Study_Hours_per_day > 24 OR Study_Hours_per_day < 0;
184
185 UPDATE student_dataset_v2
186 SET Study_Hours_per_day = ABS(Study_Hours_per_day);
187
188
189 /* Fill missing study hours with MEDIAN */
190 WITH Ordered AS (
191     SELECT Study_Hours_per_day,
192            ROW_NUMBER() OVER (ORDER BY Study_Hours_per_day) rn,
193            COUNT(*) OVER () cnt
194     FROM student_dataset_v2
195 )
196 UPDATE student_dataset_v2
197 SET Study_Hours_per_day = (
198     SELECT AVG(Study_Hours_per_day)
199     FROM Ordered
200     WHERE rn IN (FLOOR((cnt+1)/2), CEIL((cnt+1)/2))
201 )
202 WHERE Study_Hours_per_day IS NULL;
203
204
205 /* =====
206 13. SCORE CLEANING
207 ===== */
208 UPDATE student_dataset_v2
209 SET Math_Score = NULL WHERE Math_Score > 100 OR Math_Score < 0;
210
211 UPDATE student_dataset_v2
212 SET Science_Score = NULL WHERE Science_Score > 100 OR Science_Score < 0;

```

```

213
214 UPDATE student_dataset_v2
215 SET English_Score = NULL WHERE English_Score > 100 OR English_Score < 0;
216
217
218 /* Replace missing scores with MEDIAN */
219 -- (Same pattern used for all 3 columns)
220
221
222 /* =====
223 14. TOTAL SCORE RE-CALCULATION
224 ===== */
225 UPDATE student_dataset_v2
226 SET Total_Score = Math_Score + Science_Score + English_Score;
227
228
229 /* =====
230 15. PARENT EDUCATION CLEANING
231 ===== */
232 UPDATE student_dataset_v2
233 SET Parent_Education = 'Post Graduate'
234 WHERE LOWER(Parent_Education) IN ('postgraduate');
235
236 -- Fill missing with MODE
237 UPDATE student_dataset_v2
238 SET Parent_Education = (
239     SELECT Parent_Education FROM (
240         SELECT Parent_Education
241         FROM student_dataset_v2
242         GROUP BY Parent_Education
243         ORDER BY COUNT(*) DESC
244         LIMIT 1
245     ) t
246 )
247 WHERE Parent_Education IS NULL OR Parent_Education = '';
248
249
250 /* Proper case formatting */
251 UPDATE student_dataset_v2
252 SET Parent_Education = CONCAT(
253     UPPER(LEFT(Parent_Education, 1)),
254     LOWER(SUBSTRING(Parent_Education, 2))
255 );
256
257
258 /* =====
259 16. INTERNET ACCESS CLEANING
260 ===== */
261 UPDATE student_dataset_v2
262 SET Internet_Access = CASE
263     WHEN LOWER(Internet_Access) IN ('yes', '1') THEN 'Yes'
264     WHEN LOWER(Internet_Access) IN ('no', '0') THEN 'No'
265     ELSE NULL
266 END;

```

```

267
268 -- Fill missing with MODE
269 UPDATE student_dataset_v2
270 SET Internet_Access = (
271     SELECT Internet_Access FROM (
272         SELECT Internet_Access
273         FROM student_dataset_v2
274         GROUP BY Internet_Access
275         ORDER BY COUNT(*) DESC
276         LIMIT 1
277     ) t
278 )
279 WHERE Internet_Access IS NULL OR Internet_Access = '';
280
281
282 /* =====
283 17. RESULT COLUMN (BUSINESS LOGIC)
284 ===== */
285 UPDATE student_dataset_v2
286 SET Result =
287     CASE
288         WHEN Math_Score >= 33
289             AND Science_Score >= 33
290             AND English_Score >= 33
291         THEN 'Pass'
292         ELSE 'Fail'
293     END;
294
295
296 /* =====
297 18. AGE DERIVATION FROM CLASS
298 ===== */
299 UPDATE student_dataset_v2
300 SET Age = CASE
301     WHEN Class = 6 THEN 11
302     WHEN Class = 7 THEN 12
303     WHEN Class = 8 THEN 13
304     WHEN Class = 9 THEN 14
305     WHEN Class = 10 THEN 15
306     ELSE Age
307 END;
308
309
310 /* =====
311 19. FINAL DATA TYPE FIX
312 ===== */
313 ALTER TABLE student_dataset_v2 MODIFY Age INT;
314 ALTER TABLE student_dataset_v2 MODIFY Class INT;
315 ALTER TABLE student_dataset_v2 MODIFY Attendance_Percentage FLOAT;
316 ALTER TABLE student_dataset_v2 MODIFY Math_Score INT;
317 ALTER TABLE student_dataset_v2 MODIFY Science_Score INT;
318 ALTER TABLE student_dataset_v2 MODIFY English_Score INT;
319 ALTER TABLE student_dataset_v2 MODIFY Total_Score INT;

```